# CAPSTONE PROJECT: INTERIM STAGE

Project Code: XH3
Project Title:
Enterprise Data Leakage Prevention System

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# Background

- More of our data is being stored in digital form
- More convenient? Yes.
- More safe? Not really.

# Major Challenges

- Data Leakage
  - Yahoo 2014: A hacker from Russia stole the database containing 500 million users' information, and Yahoo only realized this 2 years after the incident [1]
  - Apple 2022: a bug in the Safari app led to the exposure of user's AppleIDs [2]
  - Facebook 2018: Sensitive data of 533 million users exposed to Cambridge Analytica [3]

  - —> A major problem even for large IT companies!!

# Goal of the Project

- To create a Data Leakage prevention system that prevents the leakage of a company's data and especially sensitive data, to the outer world.
- The system will be a web application, designed by setting rules on how data should be transmitted within the system, to ensure data is:
  - Being managed correctly,
  - Unable to be leaked through user's actions within the system, and:
  - Encrypted to protect it from attacks.

# Major Features

1. Data Analytics Engine
    a. Pattern Recognition: To help the system identify abnormal or suspicious activities regarding action with a file at a certain time
    b. Risk Scoring: To help users understand the risk of sharing a certain file
    c. Data Classification: To help identify files that contain sensitive data and label them, to prevent a user from sharing confidential information carelessly

# Major Features

2. Admin Privileges
   a. Admins can view all logs of all users on the system
      - To track suspicious activity
   b. Admins cannot view or have access to any files on the system
      - To prevent leakage of files due to unnecessary access
   c. Admins can change System settings and accept requests of changing account settings from other users

# Major Features

3. File Self-Destruction
   a. A "Recycle Bin" is included in the system
      - When a user deletes a file, the file first moves into the bin
   b. A 7-day-countdown is set for each file moved into the bin
      - The file can be recovered anytime within the 7-day-countdown
      - The file can also be selected to be deleted immediately and skip the 7-day-countdown, and will be completely overwritten
   c. Once the 7-day-countdown expires, the file will be completely overwritten with DoD 5220.22-M

# Major Features

4. Watermark Processing
   - Needed because files may be leaked with hardware capture

a. A watermark will be embedded in the file upon every view or edit of the file

b. The watermark will be in the format where the user viewing or editing the file and the time of action will be included

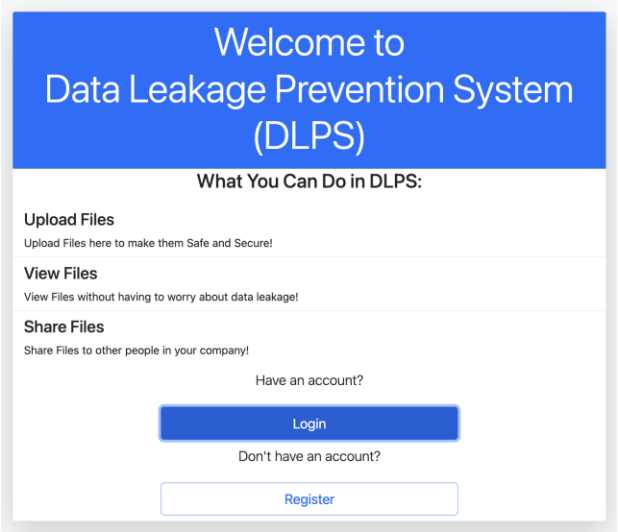
—> Help trace origin of leakage if it really happens

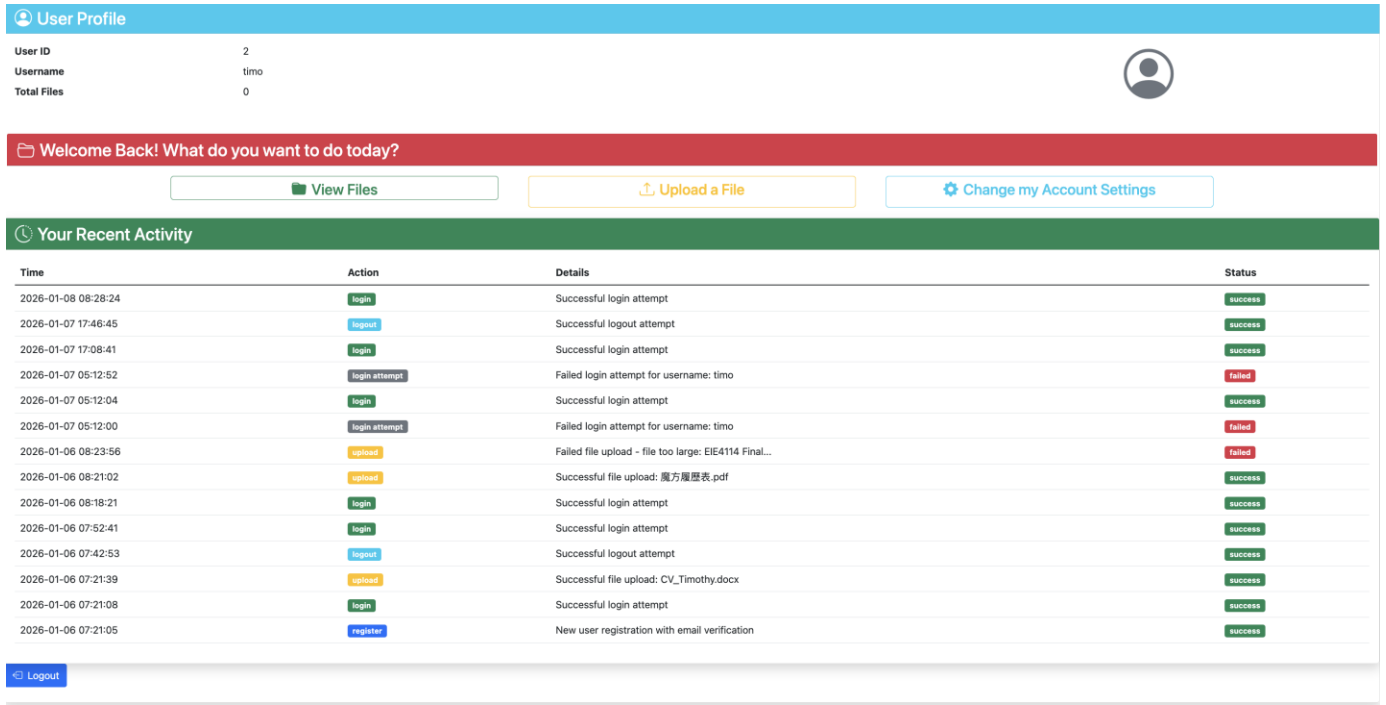
# Sourcing of Tools and Frameworks

- **Server: Python Flask**
- **UI design**: Bootstrap
- **Language**: HTML, Python, JavaScript, CSS, SQL

# UI design

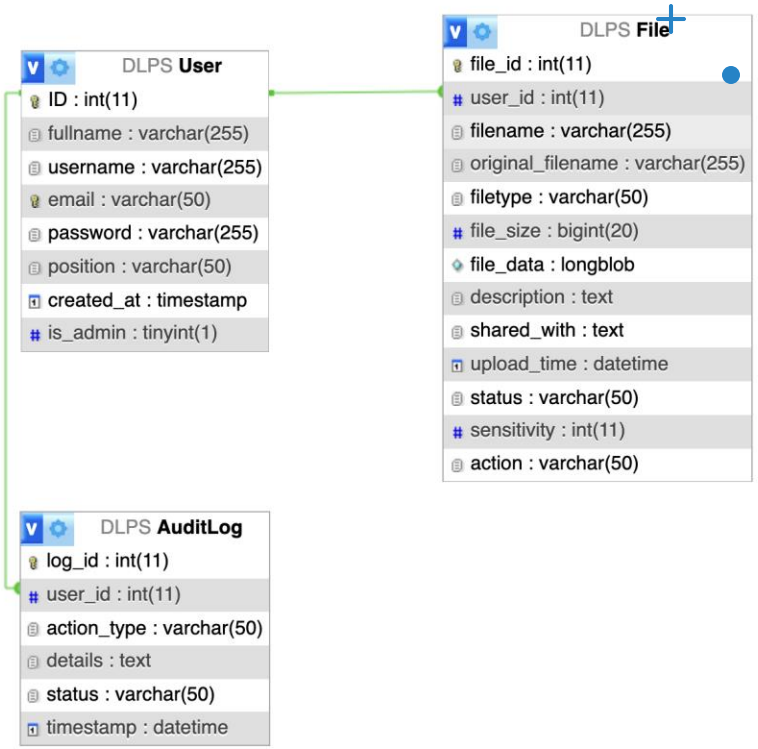
- Bootstrap: An open-source CSS Framework
- Straightforward, easy-to-understand, user-friendly



# Database design

- 3 tables for now
- **User ↔ AuditLog**
  *One-to-Many*: One user (User.ID) can be involved in multiple AuditLogs
  Foreign Key: User.ID → AuditLog.user_id
- **User ↔ File**
  **One-to-Many**: One user (User.ID) can be associated with multiple Files
  Foreign Key: User.ID → File.user_id



# Models

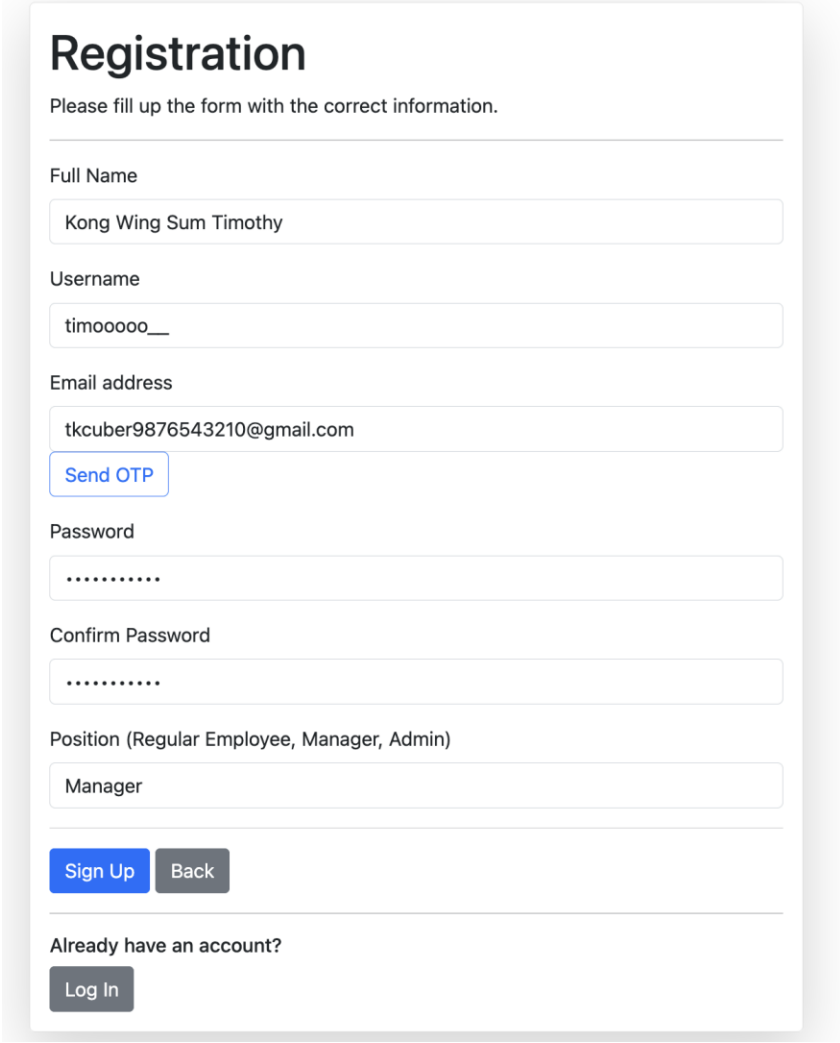
- 3 Models for now: User, File, AuditLog
- Uses variables from the Database to build the model for use in Flask
- Not finalized yet as other models are not yet implemented (e.g. Share, Bin models)

models.py > ...

```python
from flask_sqlalchemy import SQLAlchemy
from datetime import datetime
from werkzeug.security import generate_password_hash, check_password_hash
# from flask_wtf import FlaskForm
# from wtforms import StringField, PasswordField, SubmitField, SelectField
# from wtforms.validators import InputRequired, ValidationError, Length

db = SQLAlchemy()

class User(db.Model):
    __tablename__ = 'User'

    id = db.Column('ID', db.Integer, primary_key=True)
    fullname = db.Column(db.String(150), nullable=False)
    username = db.Column(db.String(80), unique=True, nullable=False)
    email = db.Column(db.String(80), unique=True, nullable=False)
    password = db.Column('password', db.String(255), nullable=False)
    #password_hash = db.Column(db.String(200))
    position = db.Column(db.String(50), nullable=False)  # Employee, Manager, Admin
    created_at = db.Column(db.DateTime, default=datetime.utcnow)
    is_admin = db.Column(db.Boolean, default=False)

    def set_password(self, password):
        self.password = generate_password_hash(password)

    def check_password(self, password):
        return check_password_hash(self.password, password)

class File(db.Model):
    file_id = db.Column(db.Integer, primary_key=True)
    user_id = db.Column(db.Integer, db.ForeignKey('User.ID'))
    filename = db.Column(db.String(200), nullable=False)
    original_filename = db.Column(db.String(200), nullable=False)
    filetype = db.Column(db.String(10), nullable=False)
    file_size = db.Column(db.Integer)
    description = db.Column(db.Text)
    shared_with = db.Column(db.String(255))
    #filepath = db.Column(db.String(500), nullable=False)
    upload_time = db.Column(db.DateTime, default=datetime.utcnow)
    status = db.Column(db.String(50), default='pending')  # pending, scanned, flagged, safe
    sensitivity = db.Column(db.Integer)  # 1-10
    action = db.Column(db.String(20))  # 'block', 'alert', 'quarantine', 'no action

class AuditLog(db.Model):
    __tablename__ = 'AuditLog'
    log_id = db.Column(db.Integer, primary_key=True)
    user_id = db.Column(db.Integer, db.ForeignKey('User.ID'))
    action_type = db.Column(db.String(50), nullable=False)  # e.g., 'upload', 'delete', 'share'
    details = db.Column(db.Text)  # Additional details about the action
    status = db.Column(db.String(50), default='success')  # success, failed
    timestamp = db.Column(db.DateTime, default=datetime.utcnow)
```

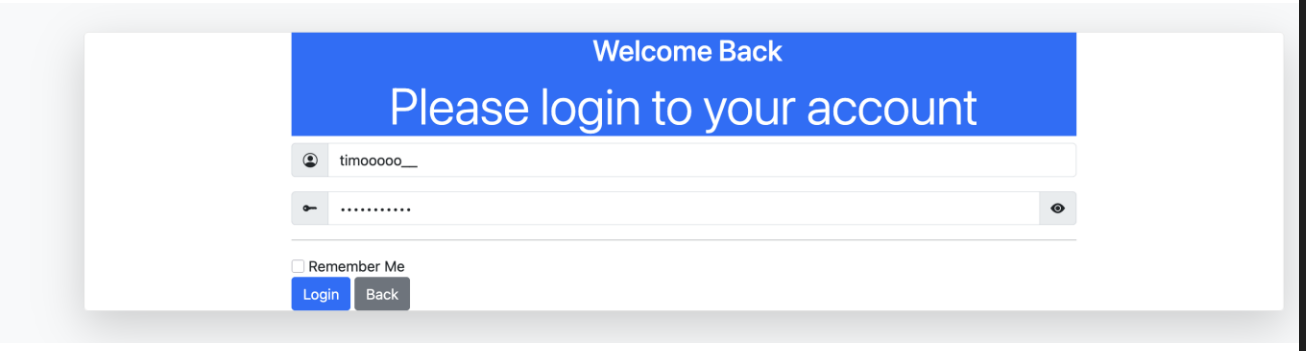
# Functions

- Register(): Intakes user's personal information, creates a User in Flask and creates a user entry in the User table in the database



# Functions

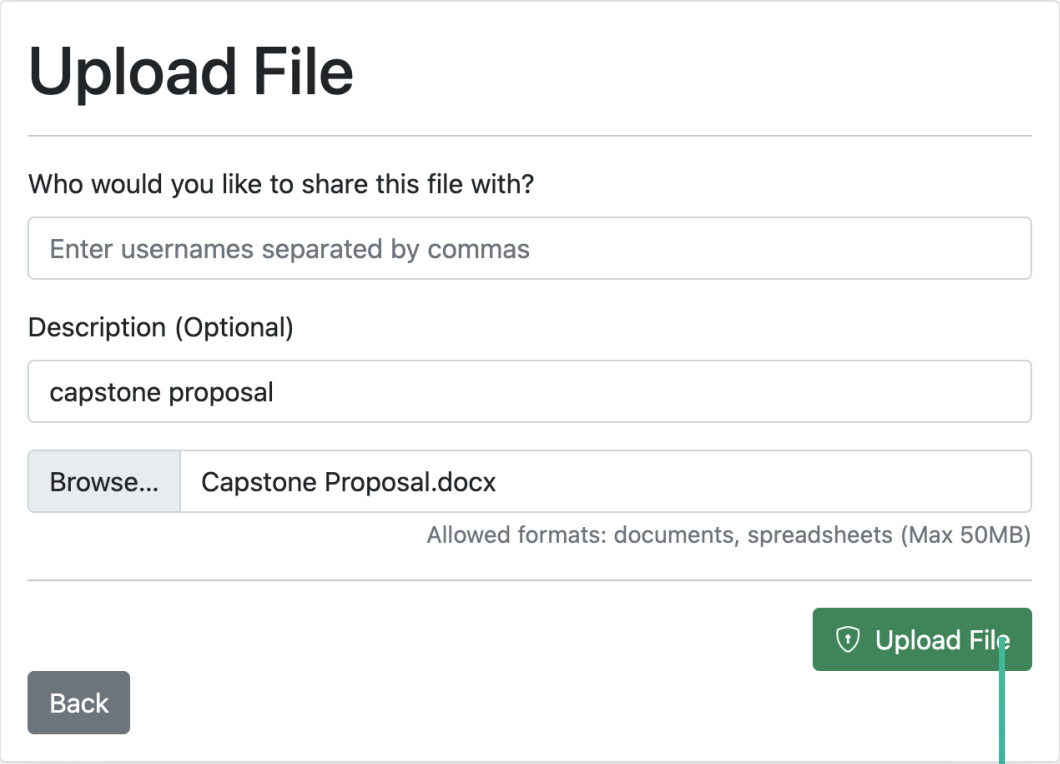
- Login(): Intakes the username and password as credentials and allows the user to access the dashboard upon login




```python
@auth.route('/login', methods=['GET', 'POST'])
def login():
    if request.method == 'POST':
        username = request.form.get('username')
        password = request.form.get('password')
        #remember = request.form.get('remember')

        if not username:
            flash('Please enter your username', 'danger')
            return render_template('login.html', username=username)
        if not password:
            flash('Please enter your password', 'danger')
            return render_template('login.html', username=username)

        user = User.query.filter_by(username=username).first()
        if not user:
            flash('Invalid username or password', 'danger')
            return render_template('login.html', username=username)

        elif user and user.check_password(password) == False:
            log = auditlog(
                user_id=user.id,
                action_type='login attempt',
                details='Failed login attempt for username: ' + username,
                status='failed'
            )
            db.session.add(log)
            db.session.commit()
            flash('Invalid username or password', 'danger')
            return render_template('login.html', username=username)

        else:
            session['user_id'] = user.id
            session['username'] = user.username
            session['is_admin'] = user.is_admin
            flash('Login successful!', 'success')
```

# Functions

- Upload(): Allows user to upload a file to the system, with the use of AES-128 Encryption method



# Functions

- AuditLog(): Logs the events made by the user into the database with the time of event, action type and status.

```python
def auditlog(user_id, action_type, details='', status='success'):
    """Helper function to log audit events"""
    try:
        log_entry = AuditLog(
            user_id=user_id,
            action_type=action_type,
            details=details,
            status=status,
            timestamp=datetime.utcnow()
        )
        db.session.add(log_entry)
        return log_entry
    except Exception as e:
        print(f"Failed to log audit event: {e}")
        db.session.rollback()
```

# AuditLog function

## User Profile

| | |
|---|---|
| User ID | 2 |
| Username | timo |
| Total Files | 0 |

## Welcome Back! What do you want to do today?

View Files | Upload a File | Change my Account Settings

## Your Recent Activity

| Time | Action | Details | Status |
|---|---|---|---|
| 2026-01-08 08:28:24 | login | Successful login attempt | success |
| 2026-01-07 17:46:45 | logout | Successful logout attempt | success |
| 2026-01-07 17:08:41 | login | Successful login attempt | success |
| 2026-01-07 05:12:52 | login attempt | Failed login attempt for username: timo | failed |
| 2026-01-07 05:12:04 | login | Successful login attempt | success |
| 2026-01-07 05:12:00 | login attempt | Failed login attempt for username: timo | failed |
| 2026-01-06 08:23:56 | upload | Failed file upload - file too large: EIE4114 Final... | failed |
| 2026-01-06 08:21:02 | upload | Successful file upload: 魔方履歷表.pdf | success |
| 2026-01-06 08:18:21 | login | Successful login attempt | success |
| 2026-01-06 07:52:41 | login | Successful login attempt | success |
| 2026-01-06 07:42:53 | logout | Successful logout attempt | success |
| 2026-01-06 07:21:39 | upload | Successful file upload: CV_Timothy.docx | success |
| 2026-01-06 07:21:08 | login | Successful login attempt | success |
| 2026-01-06 07:21:05 | register | New user registration with email verification | success |

Logout

# References


- [1] Trautman, Lawrence J. and Ormerod, Peter C, "Corporate Directors' and Officers' Cybersecurity Standard of Care: The Yahoo Data Breach," American University Law Review: Vol. 66: Iss. 5, Article 3, 2017.
- [2] L. Matthews, "A Nasty Safari Bug Could Leak Your Browsing Data And Google Account Information", Forbes, Jan. 17, 2022 [Online]. Available: https://www.forbes.com/sites/leemathews/2022/01/17/a-nasty-safari-bug-could-leak-your-browsing-data-and-google-account-information/ [Accessed Jan. 03, 2026]
- [3] N. Confessore, "Cambridge Analytica and Facebook: The Scandal and the Fallout So Far", The New York Times, Apr. 4, 2018 [Online]. Available: https://www.nytimes.com/2018/04/04/us/politics/cambridge-analytica-scandal-fallout.html [Accessed Oct. 20, 2025]

# DEMO OF THE SYSTEM